Biol 215 – Final Exam Name KEY

This exam is open book, and you are allowed to use any notes, course materials, textbooks, etc. that you wish to. However, the exam must be your own work. ***Communicating with anyone other than the instructor during the exam is grounds for an F in the class.***

1. Mammals are thought to adjust the length of their small intestines in response to the amount of food they have available to eat. Since intestines are made of living tissue, when a mammal is poorly fed the energy it expends to support excess intestinal tissue is wasted energy, and it will reduce the size of its intestine accordingly. If the food supply increases, it will grow its small intestine longer to increase its ability to absorb the nutrients in the food.

Janelle decides to test this hypothesis by feeding mice different amounts of food, and then measuring the lengths of their intestines. She randomly assigns 10 mice each to four different food treatment levels – 70% of normal, 80% of normal, 90% of normal, and 100% of normal. She feeds the mice on these diets for one month, and then sacrifices them, dissects their small intestines, empties them of food and weighs them.

Data from this experiment are in file Question1.MTW. The first column (Treatment) gives the treatment group, and the second column (SI weight) gives the weight of the small intestine, in grams.

A. What procedure will you use to compare the mean small intestine weights among these four?

Comparing four means is best done with ANOVA.

B. Test for HOV, and report the test statistic and p-value for Levene's test in the table to the right.

HOV

Levene's test stat.	*0.28*
p	*0.843*
HOV? (yes/no)	*Yes*

C. Test for normality for each of the four food treatment groups, and report the results in the table to the right. Watch the ordering – 100% will show up as the first result in MINITAB.

Normality

	70%	80%	90%	100%
AD	*0.234*	*0.334*	*0.276*	*0.298*
p	*0.721*	*0.433*	*0.574*	*0.519*
Normal? (yes/no)	*Yes*	*Yes*	*Yes*	*Yes*

D. What is the null hypothesis for this test of differences in mean small intestine weights among the food treatments?

$$H_o: \mu_{70pct} = \mu_{80pct} = \mu_{90pct} = \mu_{100pct}$$

E. Report the ANOVA table here for your test of the null hypothesis.

Source	DF	SS	MS	F	p
Treatment	*3*	*0.12*	*0.040*	*38.33*	*<0.001*
Error	*36*	*0.04*	*0.001*		
Total	*39*	*0.16*		Reject or retain Ho?	*Reject*

F. Can you tell from the ANOVA table alone which treatment groups differ from one another? Why or why not?

No, the omnibus test in the ANOVA table just tells you that there is a difference between at least one pair of means, not which means are different.

G. Report the grouping information based on your Tukey comparisons in the table to the right. Draw circles around any pairs of means that are not different from one another.

Grouping information

Trtmt.	Mean (2 dec.)	Letter
100%	*0.20*	*A*
90%	*0.13*	*B*
80%	*0.11*	*B*
70%	*0.05*	*C*

H. Should Janelle conclude that mice reduce the sizes of their small intestines when they are eating less? Explain how your results led you to this conclusion.

Yes, the means get smaller as the amount of food provided decreases, so that the mean SI weight at 100% food is significantly greater than it is at 70% food.

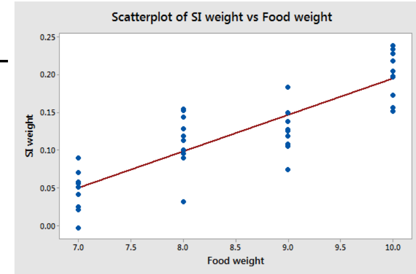
2. Janelle notices that the change in small intestine length with a change food levels looks pretty consistent on her interval plot of means, and wants to test whether it's a constant, linear relationship. The 100% food treatment refers to 10 g fed to each mouse per day, and thus 90% is 9 g, 80% is 8 g, and 70% is 7 g. She enters these weights as a numeric variable, so that she can relate the small intestine weight to the weight of food.

The data are in Question2.MTW. The SI weight data is the same, but now Food weight refers to the amount of food (in grams) given the mice each day.

A. What kind of analysis should you use to test for a linear relationship between food levels on weight of the small intestine?

Simple linear regression.

B. Make a graph that best illustrates the analysis that you plan to run – is there any reason to suspect that a straight line is a poor representation of the relationship between the variables? Why or why not?



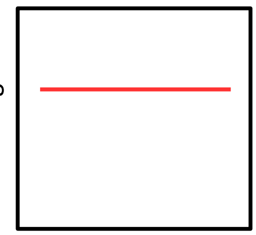
The scatterplot with fitted line shows a linear relationship between SI weight and food weight, so regression is a good choice.

C. What is the null hypothesis for this test?

$H_o\colon\beta=0$

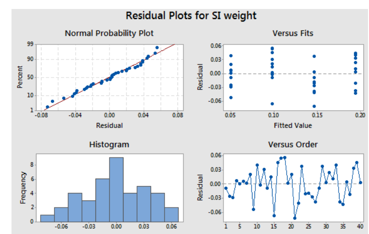
D. Sketch a graph on the blank axes to the right that shows what the null hypothesis looks like for your test of the effect of food weight on small intestine weight.

SI weight



Food weight

E. Run the analysis. Based on the residual plots, did you violate any of the assumptions of your regression analysis? How do you know if you met the normality assumption? How do you know if you met the HOV assumption?



Residual plots look good – the normal probability plot shows the residuals are normally distributed, and the residual vs. fitted plot shows equal variance from left to right.

F. Report the ANOVA table here:

Source	DF	SS	MS	F	p
Regression	*1*	*0.12*	*0.12*	*103.14*	*<0.001*
Error	*38*	*0.04*	*0.001*		
Total	*39*	*0.16*		Reject or retain Ho?	*Reject*

G. Report the regression equation here. Circle the slope, and underline the intercept.

SI weight = <u>- 0.2862</u> + 0.04807 Food weight

H. Which parameter (slope or intercept) tells you how much of a change in small intestine weight to expect if you increase the food weight by 1 g?

The slope tells us this.

I. What would we predict the small intestine weight to be if we didn't feed the mice at all (that is, food weight = 0)? For what two reasons might we not trust the accuracy of this predicted value?

This is the intercept, which is -0.282. The first reason not to trust it is that it's not possible to have a negative small intestine weight, so we know it's wrong. The second reason is that the smallest amount of food we tested was 7 g, so we don't have any data at 0 g of food, so we would be extrapolating beyond the data if we interpreted the intercept.

J. What percentage of the variation in SI weight was explained by food weight?

This is the coefficient of determination, r2, and it is equal to 73.1%.

3. A wide variety of bird species use our coastal lagoons as habitat, with over 80 species detected just in the San Dieguito River Estuary. Some species are year-round residents, such that they are about equally common all year long, and others migrate through and are only present at all in the Spring and Fall. Other species fall someplace in between – some of the individuals of the species stay in the Estuary all year, but others move through on their way to breeding grounds or wintering grounds to the north or south, respectively.

Jason is interested in whether mallards, which are a very common species of duck in the area, are one of these species that has both a permanent resident part, and a migratory part. He compiles data on the numbers of mallards detected during monthly surveys of the Estuary conducted over a five year period into a table that shows the number of mallards counted in the Winter, Spring, Summer and Fall. He wishes to test whether the frequencies are the same each season, or instead are greater during the Spring and Fall migration than they are during the Summer and Winter.

The data are in file Question3.MTW. The season is indicated in the first column, and the Frequency column has the total number of mallards observed over five years of counts within each season. The Expected proportion column gives the relative frequencies expected if mallards are not migratory at all.

A. What test should you run to determine whether the observed number of mallards matches the expected number?

A Chi-square goodness of fit test.

B. What is the null hypothesis for this test?

$$H_o : Observed = Expected$$

C. Run the analysis and present the results below.

N ___*5157*___ DF ___*3*___ Chi-square ___*159.002*___ p ___*<0.001*___

D. Do you reject the null or retain it? If you reject it, in which seasons are there more mallards than expected?

Reject the null – mallards were more common than expected in the Spring and Fall.

E. Based on the results, do mallards show some migratory behavior? Explain how you know.

Yes, because they are more common during Spring and Fall migration.

4. In addition to the seasonal, migratory patterns in bird abundances we see in the area, it is also possible for migratory patterns to change from year to year. Jason decides to test whether the distribution of frequencies among the seasons depends on the year.

File Question4.MTW gives a table of mallard observations over a two year period. Each row is a single duck; the first column gives the season it was observed, and the second column gives the year.

A. What test should you run to determine if the frequencies of mallards counted each season depends on the year?

Chi-square contingency table analysis.

B. What is the null hypothesis for this test?

$H_o : Observed = Expected$

C. Run the test, using Season as the rows, and Year as the columns. Report the results below:

Chi-square = ___159.09___ d.f. = ___3___

p = ___<0.001___ Reject Ho? *Yes*

D. Fill in the table of observed frequencies, expected frequencies, and standardized residuals to the right. The seasons should be in the same order in MINITAB as in your table, but double check before you fill it in.

BT	*Year 1*	*Year 2*
F.	Obs = *788* Exp. = *719.0* Std. res. = *2.574*	Obs = *643* Exp. = *712.0* Std. res. = *-2.587*
Sp.	Obs = *940* Exp. = *794.8* Std. res. = *5.149*	Obs = *642* Exp. = *787.2* Std. res. = *-5.174*
Su.	Obs = *479* Exp. = *563.2* Std. res. = *-3.549*	Obs = *642* Exp. = *557.8* Std. res. = *3.566*
W.	Obs = *384* Exp. = *514* Std. res. = *-5.733*	Obs = *639* Exp. = *509* Std. res. = *5.761*

E. Look at the observed numbers in Year 2, and note that they are almost identical from season to season. Compare this to the observed numbers in Year 1, and to the row totals – the counts are very different between the seasons for Year 1, and overall. If you found any significant standardized residuals in Year 2, is this a mistake? How is it possible to have significant differences from expected numbers in Year 2 when the numbers are all almost identical?

Rows: Season Columns: Year

	Year 1	Year 2	All
Fall	788	643	1431
	719.0	712.0	
	2.574	-2.587	
Spring	940	642	1582
	794.8	787.2	
	5.149	-5.174	
Summer	479	642	1121
	563.2	557.8	
	-3.549	3.566	
Winter	384	639	1023
	514.0	509.0	
	-5.733	5.761	
All	2591	2566	5157

This isn't a mistake – the expected values come from the marginal totals for each season (the "All" column), which show more mallards in the fall and spring than in the summer and winter. This seasonal variation is the pattern that the expected values are based on. This general seasonal pattern is present in Year 1, but it is even more pronounced during Year 1 than in the marginal totals, so there are more mallards than expected in Fall and Spring than expected, and fewer than expected in Summer and Winter. In Year 2 the birds stopped showing a seasonal pattern at all, so when we compare them to the expected values based on the marginal totals there are fewer than expected in Fall and Spring because they don't increase as expected then. Likewise, there are more mallards than expected in Summer and Winter, because they don't decrease as expected then.

F. Based on your results, do mallard frequencies among the seasons change between the years? Use the observed frequencies and standardized residuals to explain how the frequencies are changing.

Yes, they are very seasonal in Year 1, with significantly more mallards than expected in the Fall and Spring than expected and fewer in the Summer and Winter than expected, but they become non-seasonal in Year 2. Since the marginal totals show a seasonal pattern, the lack of seasonality in Year 2 is seen as having fewer than expected in the Fall and Spring, and more than expected in Summer and Winter, but the observed counts are all between 639 and 643, indicating that the birds stopped being seasonal in the second year.